

Distant Mid-Air Interaction via Accurate Gaze and Hand Posture Estimation with an Ultrafast Pan-Tilt Camera

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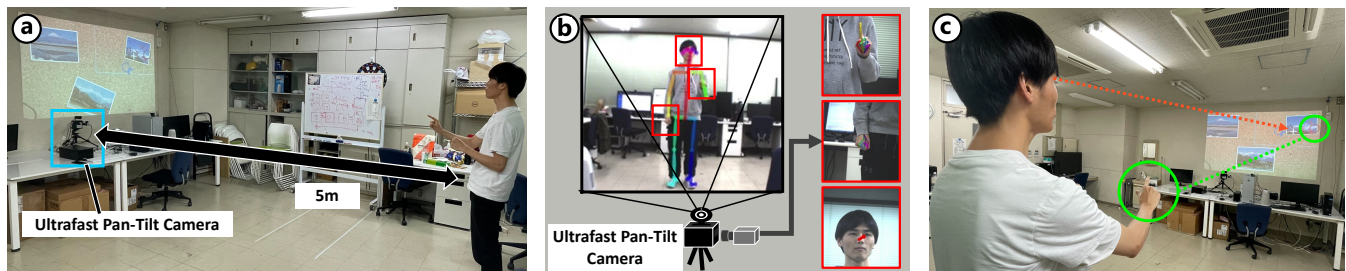


Figure 1: Distant mid-air interaction. (a) The user gazes at the screen and moves their hands 5 m away from the camera system. (b) The ultrafast pan-tilt camera simultaneously captures high-resolution images of the user's face, left hand, and right hand individually, enabling accurate estimation of gaze direction and hand posture. (c) While switching the region of interest (ROI) to the upper-right area of the large display using their gaze, the user is performing multi-touch interactions within the ROI, maintaining a natural posture without large arm movements.

Abstract

We propose a distant mid-air interaction system that uses an ultrafast pan-tilt camera to enable accurate selection and manipulation of objects across the entire large display, without large arm movements. By simultaneously capturing high-resolution images of the user's face and both hands using the ultrafast pan-tilt camera, we achieve the same level of gaze and hand posture estimation accuracy at 5 m away from the camera as that obtained at close range. In this system, the screen is divided into multiple regions, and one is selected as the region of interest (ROI) based on the gaze direction. This allows the user to perform interactions within the ROI with only slight hand movements. Our demonstration showed that users can perform multi-touch interactions across the entire large display while maintaining a natural posture with their hands in front of their chest.

CCS Concepts

• Human-centered computing → Gestural input; Touch screens.

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active sensing, gaze input, bare hand interaction

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1 Introduction

Recent advancements in deep learning have led to significant interest in distant mid-air interaction, a technology that uses sensors to recognize gestures, enabling contactless operation from a distance [5, 8]. However, when the interaction space is large, such as with a large display, adjusting the control-display (C-D) gain involves a trade-off [7]. A higher C-D gain reflects even small body movements in the operation, enabling users to cover the entire display with minimal motion, although it makes precise tasks like selecting and moving small targets more difficult.

To mitigate this trade-off, distant mid-air interaction systems have been proposed in which a region of interest (ROI) is activated by gaze and operations within it are performed using hand gestures [6, 9]. This approach allows a user to accurately select and manipulate objects across the entire ROI with minimal arm movement. Furthermore, by switching the ROI position with their gaze, they

can accurately perform these operations across the entire large display without large arm movements. However, due to the limited range in which sensors can acquire subtle eye and fingertip movements with sufficient accuracy, interactions often rely on head direction for gaze and explicit changes in hand shape. Consequently, complex operations using slight finger movements are difficult to achieve, and the gaze position on the screen tends to be unstable, making it challenging to accurately select and manipulate desired objects.

In this paper, we propose a distant mid-air interaction system that uses an ultrafast pan-tilt camera to accurately estimate gaze and both hand postures, thereby enabling accurate selection and manipulation of objects across the entire large display, without large arm movements. In this system, the screen is divided into multiple regions, and one is selected as the ROI based on the gaze direction. This allows the user to perform interactions within the ROI with only slight hand movements.

2 Distant Mid-Air Interaction System

2.1 Interaction Design

Figure 2 shows an overview of the interaction system. This system extends the mid-air multi-touch interface [2], and integrates accurate gaze estimation, thereby enabling distant mid-air interaction capable of accurate selection and manipulation of objects across the entire large display, without the need for large arm movements.

In this system, the screen is divided into multiple regions, and one of them is selected as the ROI based on the user's gaze direction. By associating the center of the ROI with the user's chest position, the user can perform multi-touch interactions within the current ROI with only slight hand movements in front of their chest.

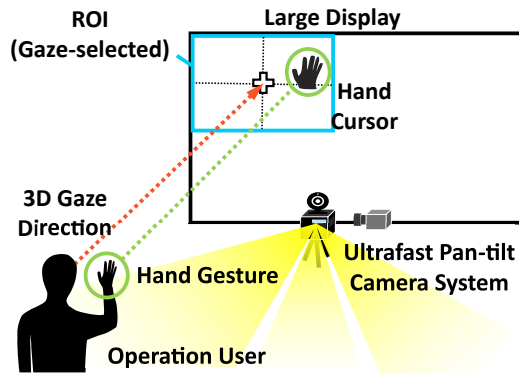


Figure 2: Overview of the distant mid-air interaction system using gaze and hand movements

2.2 Accurate Gaze and Hand Posture Estimation with an Ultrafast Pan-Tilt Camera

The camera system used in this interaction system consists of a wide-view camera and an ultrafast pan-tilt camera. The ultrafast pan-tilt camera can rapidly switch its optical axis direction within 2 milliseconds over a maximum range of ± 10 degrees horizontally and vertically, by driving only lightweight pan-tilt mirrors mounted in

front of the optical axis [4]. In this camera system, the ultrafast pan-tilt camera is equipped with a lens with a longer focal length than the wide-view camera, enabling high-spatial-resolution capture of objects detected by the wide-view camera.

In this method, first, the positions of the user's face and hands within the wide-view camera's images are identified using a full-body skeleton detector. Next, the pan-tilt mirror angles are calculated based on these identified positions. Using these mirror angles, the ultrafast pan-tilt camera rapidly adjusts its viewing direction to simultaneously capture high-spatial-resolution images of the user's face, left hand, and right hand individually. Finally, these high-spatial-resolution images are used as input for 3D gaze direction estimation and hand posture estimation. OpenPose [3] is adopted for full-body and both hands skeleton detection, while L2CS-Net [1] is adopted for the gaze estimation network. By simultaneously capturing high-resolution images of the user's face and both hands using the ultrafast pan-tilt camera, we achieve the same level of gaze and hand posture estimation accuracy at 5 m away from the camera as that obtained at close range.

3 Demonstration of the Interaction System

We conducted a demonstration of the interaction system using an implemented photo layout application. In the photo layout application, multiple photos are automatically placed on the screen, and the user can freely arrange each photo. The user can move a photo by dragging it with one fingertip, with the photo following the dragged fingertip. By using more than one finger, the user can also zoom in, zoom out, or rotate the touched photo.

Figure 3 shows the demonstration of the interaction system using the photo layout application. Regardless of the position on the display, by switching the ROI position using gaze, the user was able to precisely adjust photos by only slight movements of their fingertips while maintaining a natural posture in front of their chest. It was demonstrated that by using gaze, the slight and complex fingertip movements could be directly reflected in the operation, achieving multi-touch interaction for accurately selecting and manipulating objects across the entire display. Furthermore, it has been confirmed that these accurate operations could be consistently performed while maintaining a natural posture, making it less likely to cause fatigue or unnatural postures and reducing physical burden even during prolonged standing work. Additionally, the user was able to quickly move photos from the right edge to the left edge of the display by touching a photo and then switching the ROI with their gaze, without large arm movements. For such long-distance drag operations, this result confirmed that using gaze for distant mid-air interaction can shorten operation time and significantly reduce arm movement.

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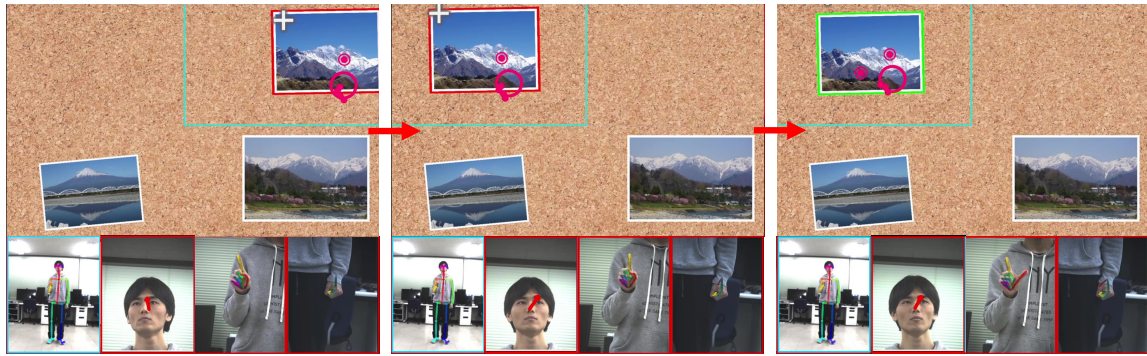


Figure 3: Demonstration of the interaction system with photo layout application. Below the application, the four images show a wide-view camera image with full-body skeleton, and three ultrafast pan-tilt camera images with gaze and both hands skeleton.

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